Driveshare Report

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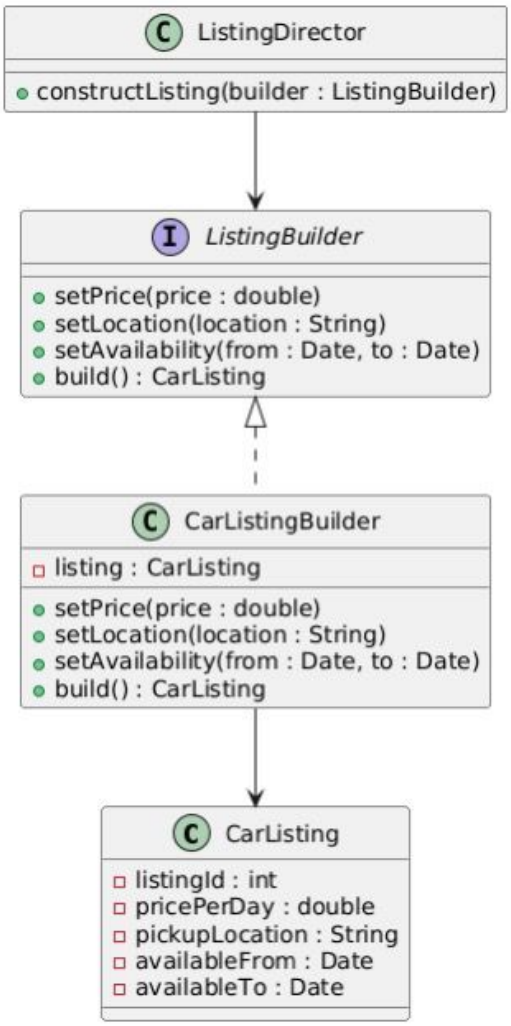
CIS 476

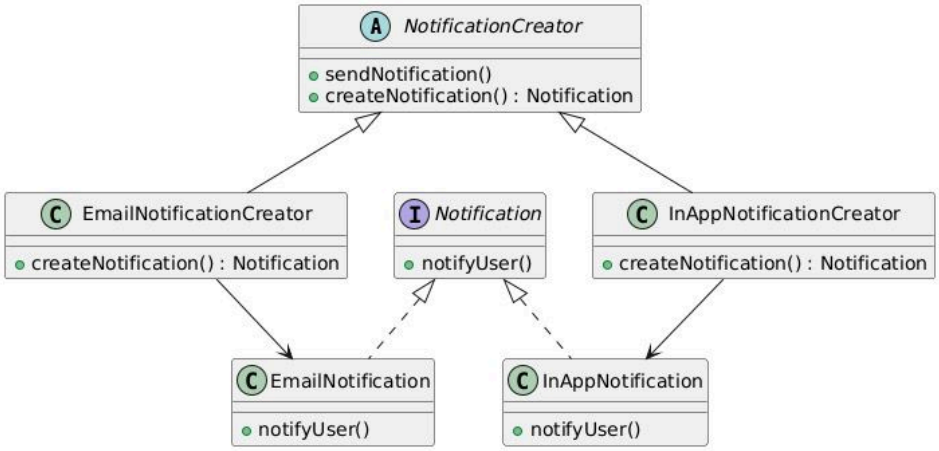
Github:
https://github.com/Hussain-Albaghdady/CIS476_TERM_PROJ.git

Video presentation URL:
https://youtu.be/WPmWgtr3cZs

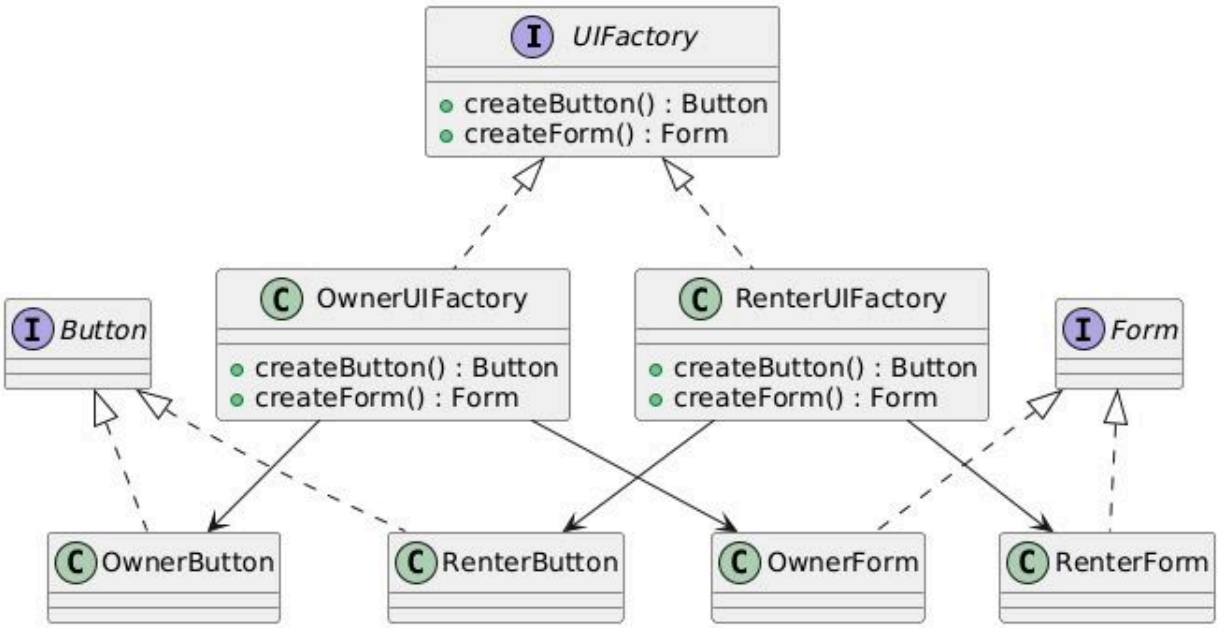
Diagrams:
1st - A builder pattern which orchestrates a listing builder interface and implements a builder for cars, it also has a step by step in doing the price, location, and availability



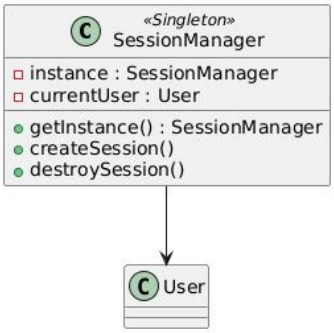
2nd-
A factory pattern for producing email notifications.



3rd- A ui factory to produce role-specific UI components

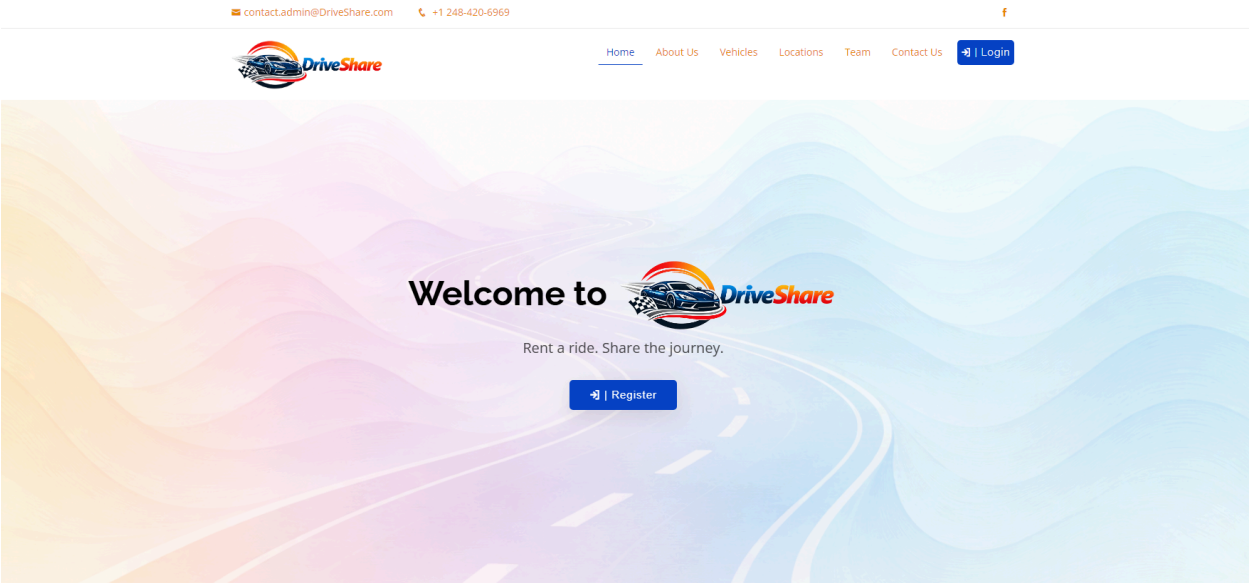


4th- A singleton pattern to ensure only one instance exists app-wide for the user.

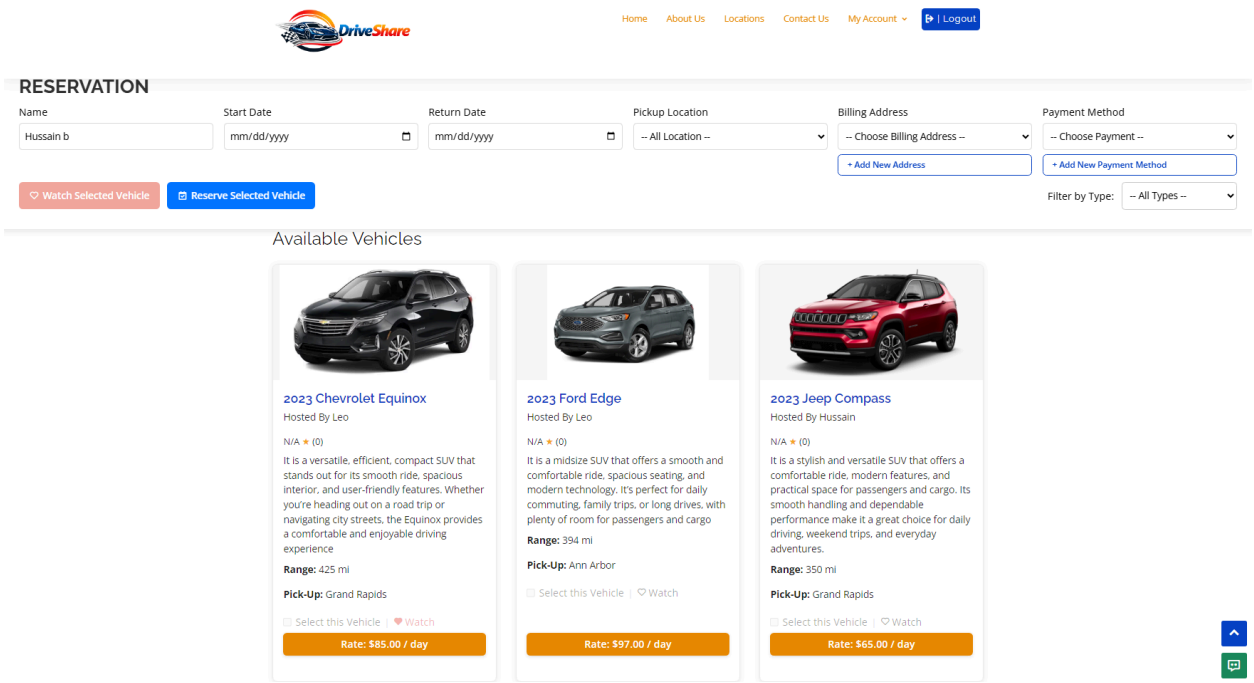


User Interface ScreenShots:

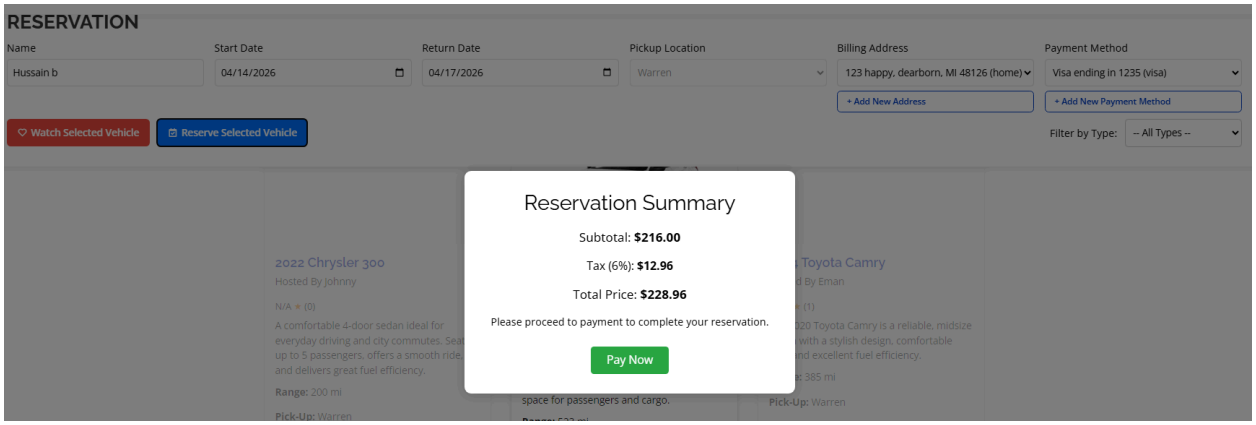
Homepage



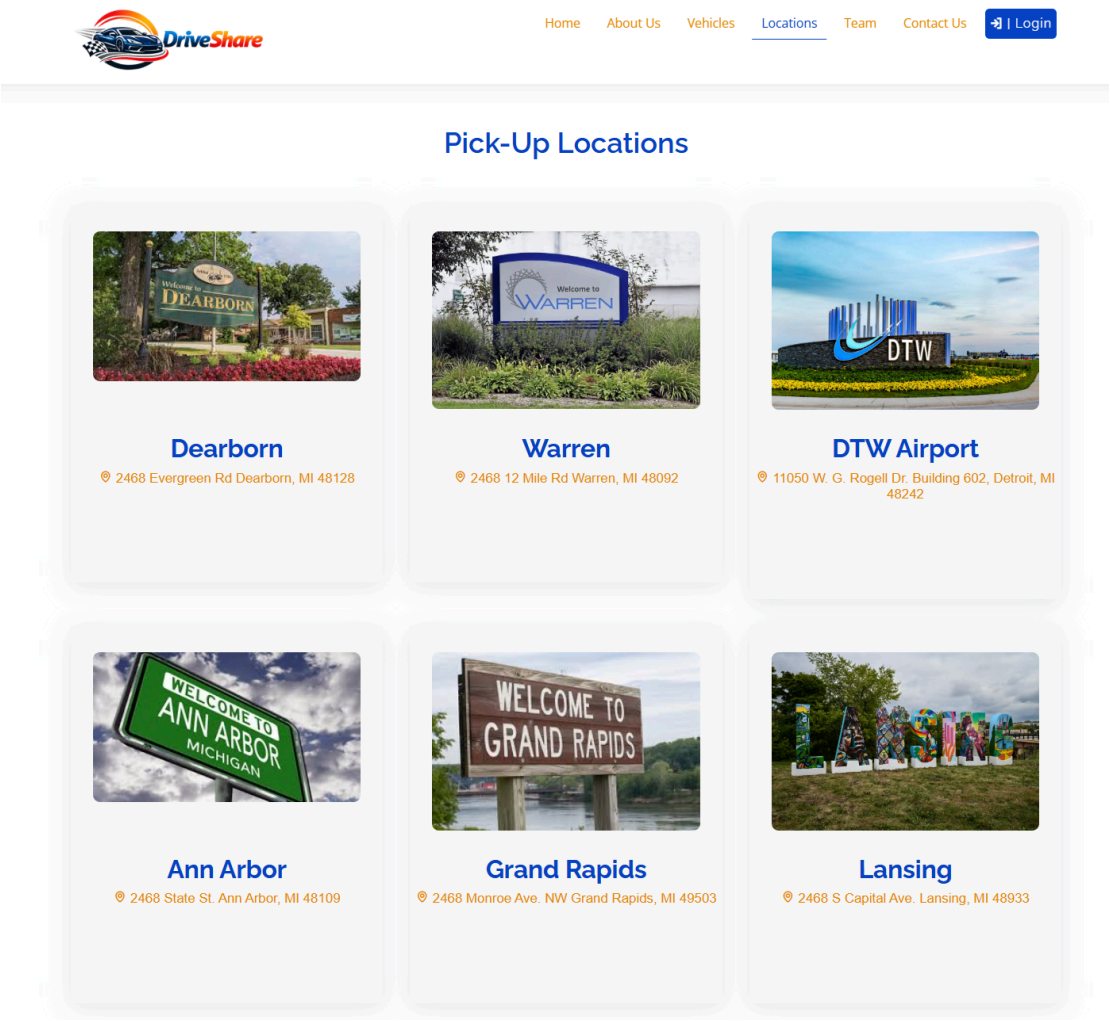
The reservation page where you select the vehicle and enter the required information to reserve it.



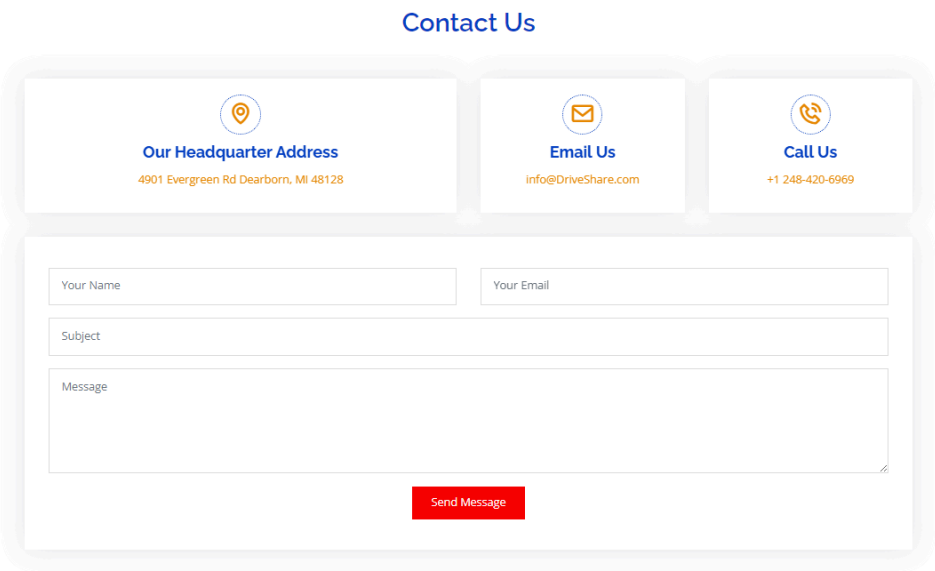
The payment pops up when you reserve a vehicle.



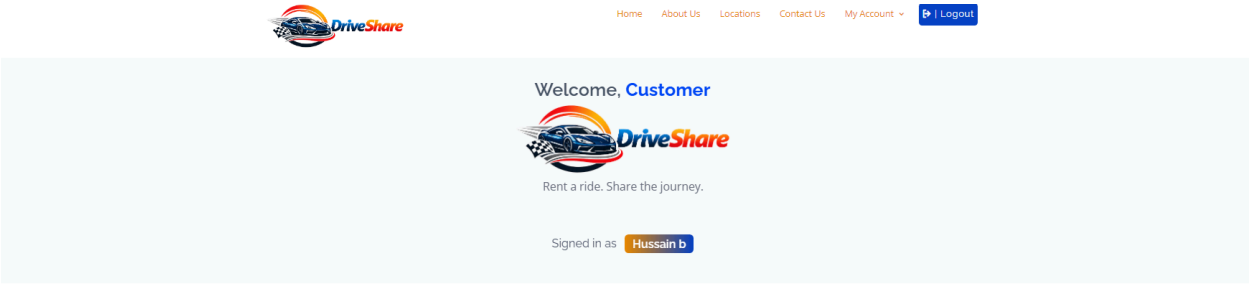
The location where the vehicles can be picked up and dropped off.



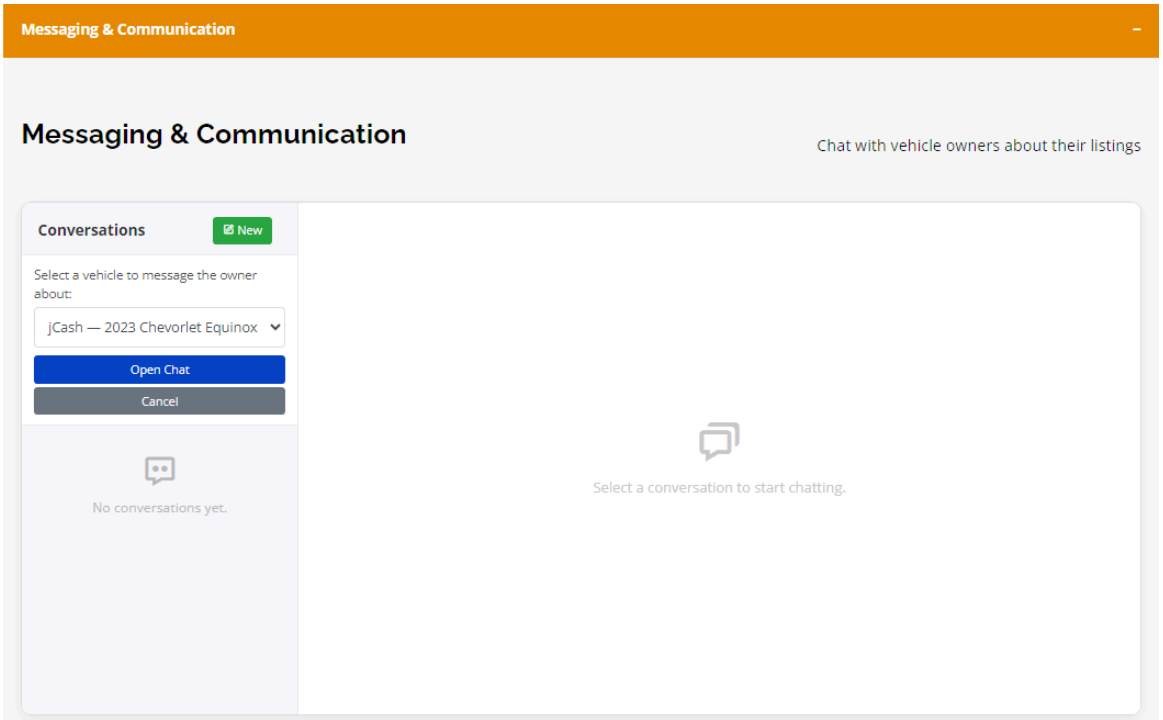
Contact information and a contact form for any concerns or questions.



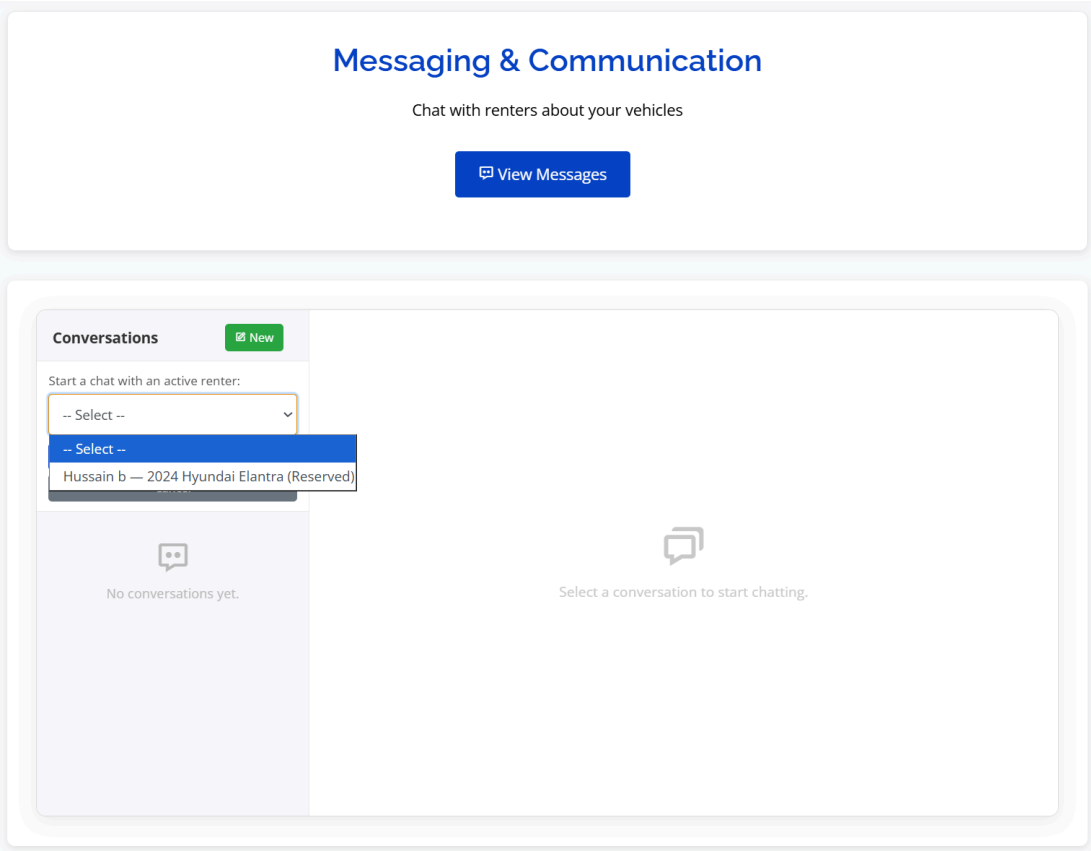
The customer dashboard with each section of information



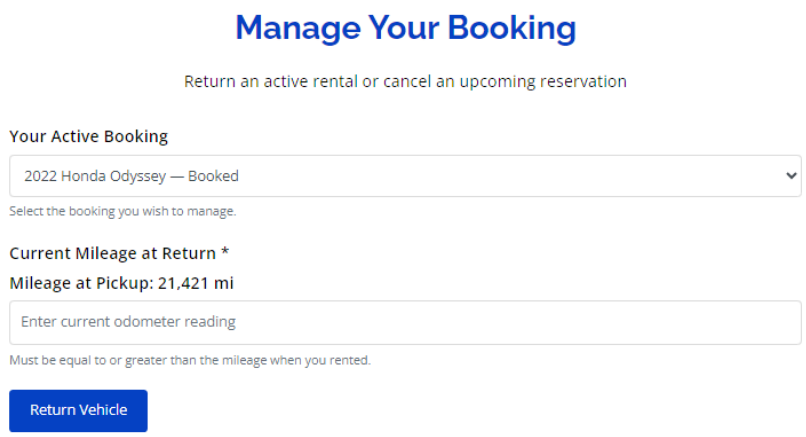
The message system on the renter side. Allow the user to start a chat with any host by vehicle.


Messaging & Communication
-
Messaging & Communication
Chat with vehicle owners about their listings
Conversations
New
Select a vehicle to message the owner about:
jCash — 2023 Chevorlet Equinox
Open Chat
Cancel
No conversations yet.
Select a conversation to start chatting.

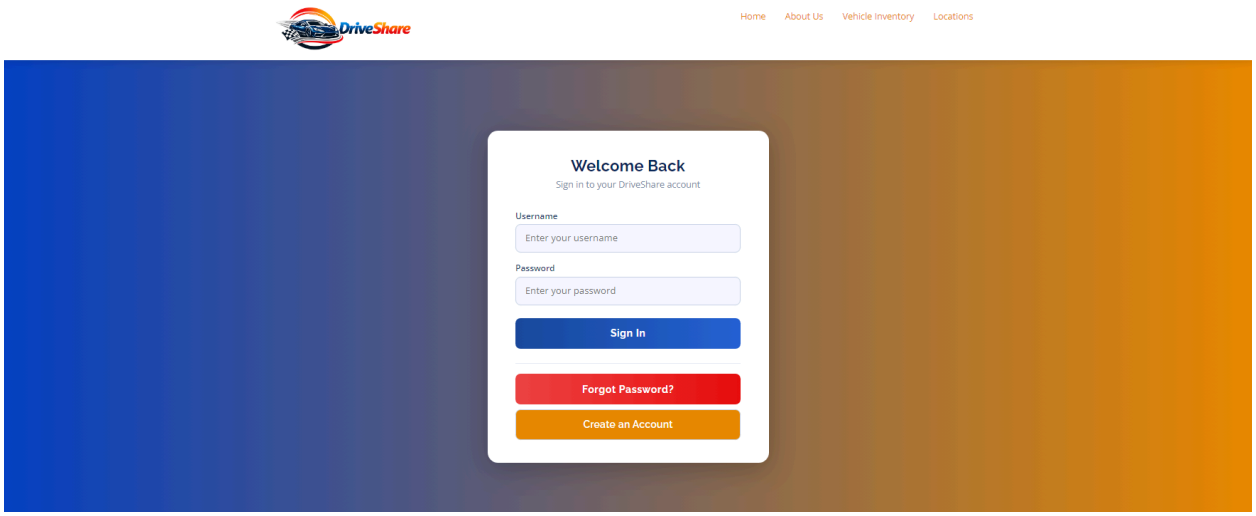
The message system on the host side. Allow the user to start a chat with renters that has a reservation for your vehicle.



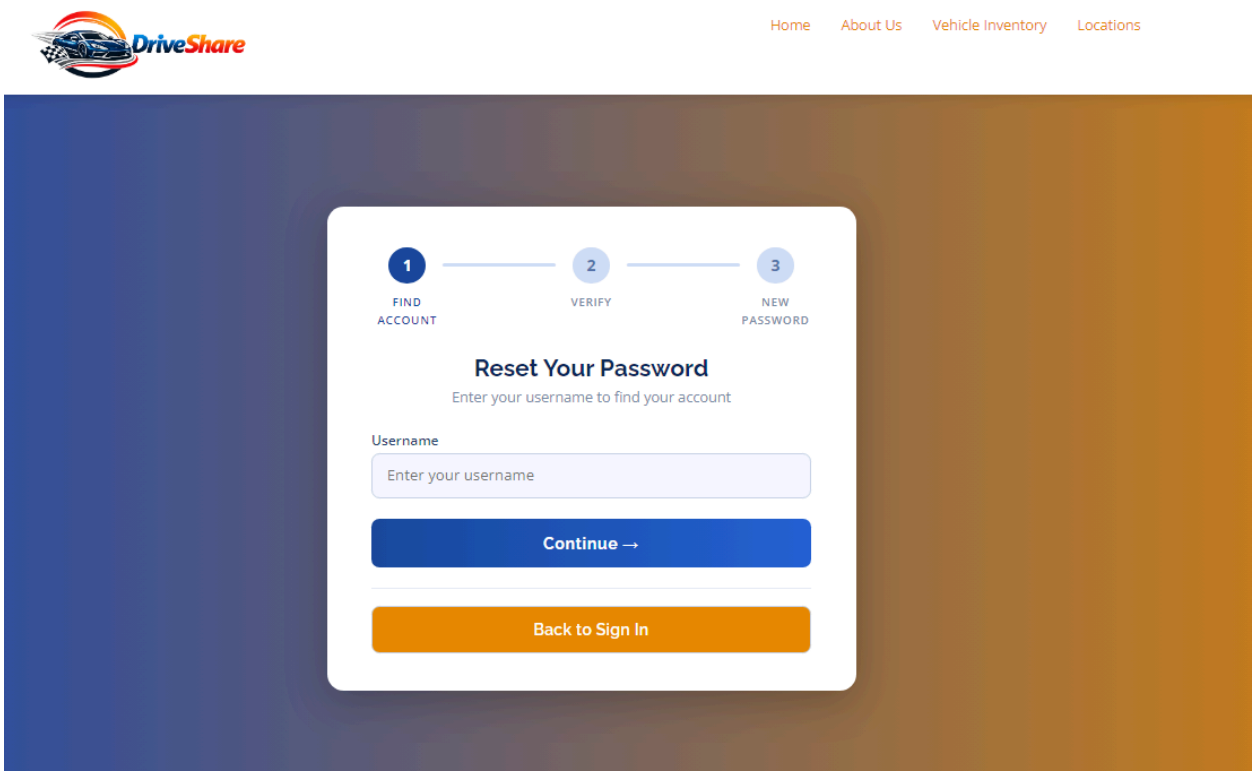
This section allows the user to return their vehicle or cancel their future reservation.



The login page form to access the host page or the renter page.



The password reset form.



The vehicles that are on your watchlist

My Watchlist

Vehicles You're Watching

Vehicle	Pick Up	Drop Off	Rate/Day	Target Price	Availability	Actions
2023 Chevrolet Equinox	2026-04-13	2026-04-16	$85.00	—	Available	Unwatch

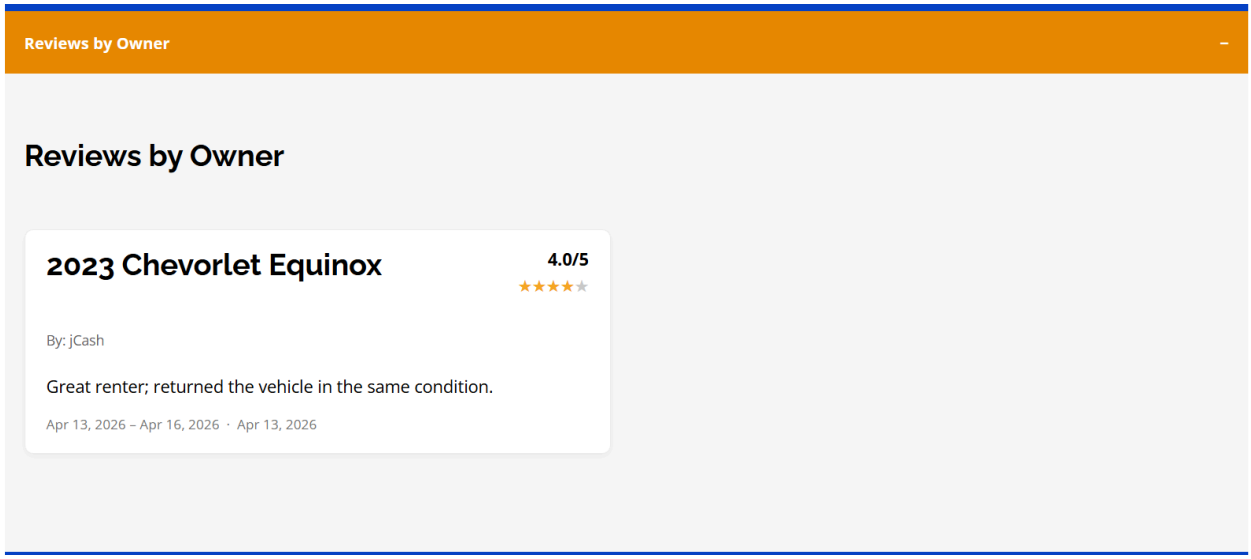
The history and current orders for users

All Reservation

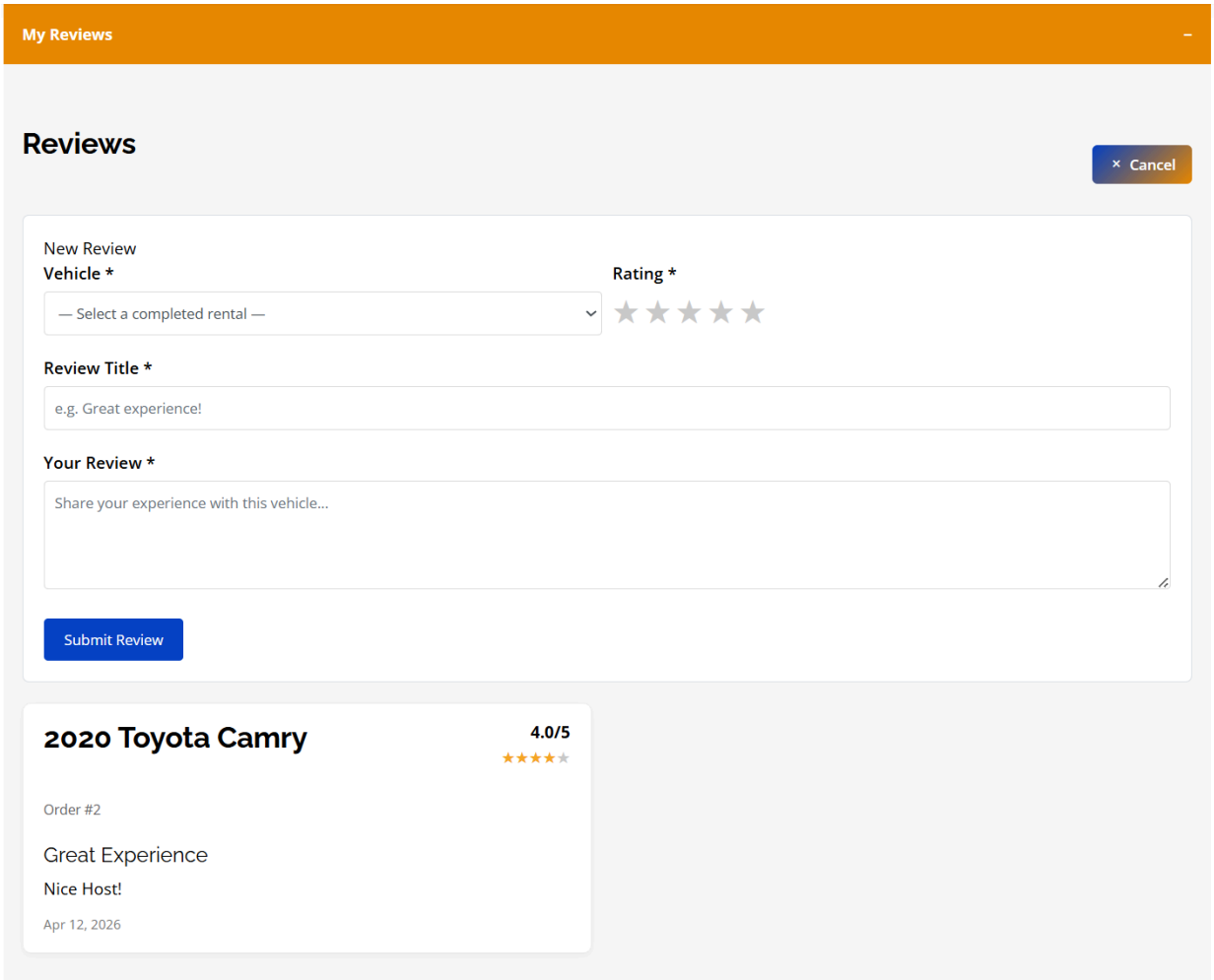
Your Latest Reservation

OrderID	Start Date	Return Date	Status	Vehicle	Mileage	Pick-Up Location	Total
1	Apr 12, 2026	Apr 24, 2026	Cancelled	2024 Jeep Wrangler 4xE	0 mi	DTW Airport	$1226.42
2	Apr 11, 2026	Apr 21, 2026	Complete	2024 Toyota Camry	1,000 mi	Warren	$524.70
3	Apr 11, 2026	Apr 12, 2026	Complete	2023 Chevorlet Equinox	30 mi	DTW Airport	$201.40
4	Apr 13, 2026	Apr 16, 2026	Complete	2023 Chevorlet Equinox	200 mi	DTW Airport	$402.80

The reviews of customer reservations by the host



The customer review of the reservation



The saved payments on a customer's dashboard. Also, adding new payment cards.

Payment Saved

Payments

+ Add Payment

Last 4 Digits	Name	Address	Expiration	Status	Card Type	Payment Nickname	Actions
2659	Meaad Hassan	N/A	09/29	active	Visa	9/11 Visa	Delete
5445	Meaad Hassan	N/A	12/98	Active	Visa	Destiny	Delete
4156	Meaad Hassan	Home	12/31	Active	Mastercard	Personal	Delete
8610	Meaad Hassan	Work	11/30	Active	Discover	word	Delete

The saved addresses on a customer's dashboard. Also, adding new addresses.

Address Saved

Address

+ Add Address

Dearborn

Delete

Meaad Hassan
3815 Evergreen Rd Dearborn, MI 48128
248-302-4848

Work

Delete

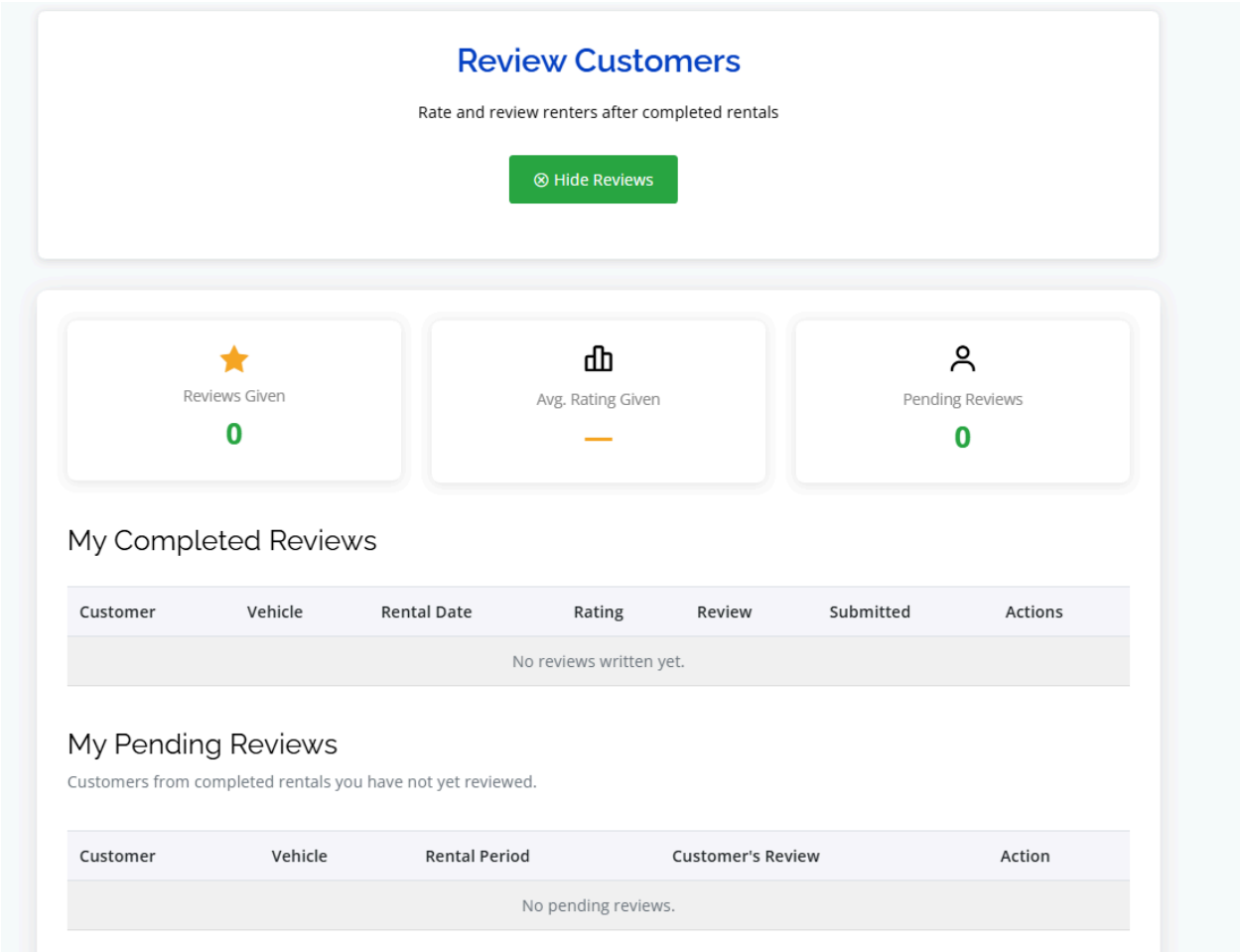
Meaad Hassan
1950 E 12 mile Road Warren, MI 48092
248-302-4848

Home

Delete

Meaad Hassan
2845 Dequindre rd Troy, MI 48083
248-525-8584

The host review section



The vehicle management of existing vehicles in the host fleet

Vehicle Management

Add new vehicles or Manage your vehicle inventory

+ Add New Vehicle

Manage Existing Vehicle

Manage Existing Vehicle Inventory

Year	Make	Model	Vehicle Class	Mileage	Range (mi)	Rate/Day	Status	Actions
2023	Chevrolet	Equinox	Compact SUV	120,000 mi	425 mi	$85.00	Available	Mark Unavailable Edit Delete
2024	Hyundai	Tucson	Compact SUV	19,000 mi	415 mi	$130.00	Available	Mark Unavailable Edit Delete
2023	Ford	Edge	Compact SUV	36,454 mi	394 mi	$97.00	Available	Mark Unavailable Edit Delete
2024	Volkswagen	Jetta	Sedan	65,245 mi	515 mi	$35.00	Available	Mark Unavailable Edit Delete
2024	Hyundai	Elantra	Sedan	46,821 mi	523 mi	$54.00	Available	Mark Unavailable Edit Delete
2023	Chevorlet	Malibu	Sedan	184,000 mi	355 mi	$40.00	Available	Mark Unavailable Edit Delete
2024	Chevorlet	Surburban	SUV	111,220 mi	400 mi	$180.00	Available	Mark Unavailable Edit Delete

Database Schemas:

Host User Schema

```
HostUsers(
  _id: ObjectId,
  hostId: int,
  fname: string,
  lname: string,
  email: string,
  username: string,
  password: string,
  user_type: string,
  created_at: string,
  updated_at: string,
  security_questions: [
    {
      question: string,
      answer: string
    }
  ],
  address: [ {} ],
  payment: [ {} ]
)
```

This collects and stores accounts information for car owners who list vehicles on DriveShare

ConatactForm Schema

```
ContactForm(
  _id: ObjectId,
  name: string,
  email: string,
  subject: string,
  message: string
)
```

This collection stores the information from users submitted by users through DriveShare

AdminUsers Schema

```
AdminUsers(
  _id: ObjectId,
  adminId: int,
  fname: string,
  lname: string,
  email: string,
  username: string,
  password: string,
  user_type: string,
  created_at: string,
  updated_at: string,
  security_questions: [
    {
      question: string,
      answer: string
    }
  ]
)
```

A collection that stores account information for DriveShare admins who manage the platform

Notfications Schema

```
Notifications(
  _id: ObjectId,
  username: string,
  type: string,
  message: string,
  link: string,
  read: boolean,
  vehicle_id: ObjectId,
  reservation_id: ObjectId,
  createdAt: Date
)
```

This is a collection for getting notifications from in-app alerts, and they are sent to users when watched vehicles become available or when prices drop

RetnalUsers Schema

```
RentalUsers(
  _id: ObjectId,
  userId: int,
  fname: string,
  lname: string,
  email: string,
  username: string,
  password: string,
  user_type: string,
  created_at: string,
  updated_at: string,
  security_questions: [
    {
      question: string,
      answer: string
    }
  ],
  address: [
    {
      customer_name: string,
```

```
      address_line1: string,
      city: string,
      state: string,
      zip_code: string,
      phone_number: string,
      address_nickname: string,
      added_at: string
    }
  ],
  payment: [
    {
      customer_name: string,
      last4: string,
      card_type: string,
      expiration: string,
      payment_address: string,
      payment_zip_code: string,
      payment_nickname: string,
      status: string,
      added_at: string
    }
  ]
)
```

This is for users on the platform who want to rent. This schema includes account information who want to rent and book vehicles on DriveShare

WatchList Schema

```
WatchList(
  _id: ObjectId,
  username: string,
  vehicle_id: ObjectId,
  target_price: null,
  start_date: string,
  end_date: string,
  created_at: Date
)
```

This is a schema to store vehicles that renters are monitoring for availability or price drops

Counters Schema

```
counters(
  _id: string,
  sequence_value: int
)
```

This is to track a counter ID that increases for users, hosts, orders and admins

Vehicles Schema

```
Vehicles(
  _id: ObjectId,
  host_username: string,
  year: int,
  make: string,
  model: string,
  category: string,
  description: string,
  mileage: int,
  range: int,
  pickup_location: string,
  rental_rate_per_day: int,
  quantity_available: int,
  availability: boolean,
  image_url: string,
  unavailable_until: Date,
  created_at: string,
  updated_at: string
)
```

A collection to store all car listings created by hosts, including details, pricing, and availability.

Reviews Schema

```
Reviews(
  _id: ObjectId,
  username: string,
  customer_name: string,
  host_username: string,
```

```
  user_id: ObjectId,
  vehicle_id: ObjectId,
  vehicle_name: string,
  reservation_id: ObjectId,
  order_id: int,
  title: string,
  body: string,
  rating: int,
  created_at: string,
  updated_at: string
)
```

A schema that stores rating and writted feedback left by renters for vehicles and hosts after a completed rental

Reservations Schema

```
Reservations(
  _id: ObjectId,
  orderId: int,
  user_id: ObjectId,
  customer_name: string,
  hostUsername: string,
  history_vehicle_id: ObjectId,
  vehicle_id: ObjectId | null,
  start_date: string,
  end_date: string,
  order_date: string,
  location: string,
  address: string,
  payment: string,
  status: string,
  total_cost: double,
  mileage: int,
  return_mileage: int,
  created_at: string,
  updated_at: string
)
```

A schema that stores vehicle booking records, including dates, payments, status, and mileage

HostReviews Schema

```
 HostReviews(
  _id: ObjectId,
  host_username: string,
  customer_username: string,
  customer_name: string,
  vehicle_id: ObjectId,
  vehicle_name: string,
  reservation_id: ObjectId,
  rating: int,
  comment: string,
  start_date: string,
  end_date: string,
  created_at: string,
  updated_at: string
)
```

A collection that stores ratings and comments submitted by hosts about renters after a completed rental

Messages Schema

```
Messages(
  _id: ObjectId,
  from_username: string,
  to_username: string,
  subject: string,
  body: string,
  vehicle_id: ObjectId,
  read: boolean,
  createdAt: Date
)
```

It is a collection of messages sent between renters and hosts for a specific vehicle